Signal Decomposition Methods for Urban Traffic Flow Analysis

A Computational Approach

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Abstract

This working paper demonstrates the ETH IVT report template within the Inkwell extension. We present computational examples that generate figures directly from markdown, combining mathematical analysis with reproducible output. The single-column layout, title page, and bibliography are all produced from YAML frontmatter and Pandoc compilation.

Keywords
traffic simulation; Fourier analysis; reproducible research

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1 Introduction

Transport simulation requires precise signal decomposition methods to analyze periodic traffic flow patterns. This working paper demonstrates the ETH IVT report template within the Inkwell extension, combining computational examples with academic writing.

The literate programming paradigm [1] allows code and prose to coexist. Inkwell extends this to compiled PDF output: code blocks execute, and their results (figures, tables, text) appear in the final document. Pandoc [2] handles the conversion from markdown to LaTeX.

2 Mathematical Framework

The Fourier partial sum approximating a square wave is given by Equation 1:

$$f_n(x) = \sum_{k=1}^{n} \frac{4}{(2k-1)\pi} \sin\bigl((2k-1)x\bigr) \tag{1}$$

As $n \to \infty$, the partial sums converge pointwise to the square wave at all points of continuity. The overshoot near discontinuities is the GIBBS PHENOMENON, which persists at approximately 9% of the jump regardless of the number of terms.

3 Computational Results

Figure **??** shows the Fourier partial sums converging to the square wave as n increases.

```
{python file=".inkwell/scripts/sine_plot.py" output="sine_plot" caption="Fourier
partial sums converging to a square wave." label="fourier"}
```

Figure **??** presents a simulated scatter plot with linear regression, demonstrating the integration of Python-generated figures.

```
{python file=".inkwell/scripts/scatter.py" output="scatter" caption="Scatter plot
with linear regression." label="scatter"}
```

4 Discussion

The examples in Section 3 demonstrate that Inkwell produces publication-quality figures from Python scripts. Equation 1 in Section 2 confirms that LaTeX math compiles correctly within the ETH report layout. The single-column format with the IVT title page provides a clean working paper presentation.

NumPy [3] and Matplotlib [4] provide the computational foundation for the visualizations shown in Figure **??** and Figure **??**.

References

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